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Novelty Must Gain Novelty

Entropy, rumination, and why an app that never ends cannot close a loop

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With drafting, literature verification, and research assistance from Grok Bot

> **Conceptual, not validated science.** This note restates a personal postulate, checks it against published research, and refines it into a testable design argument. It is not a discovered law of the brain, a diagnostic instrument, or evidence that Project Ariel treats rumination, depression, or compulsive use.

Abstract

Ryan Espinosa postulated that novelty is entropy consumed as uncertainty is lost; that humans can reduce entropy; and that if the release of novelty does not itself gain novelty—if no new connections form among thoughts—those thoughts will ruminate, the rumination may not stop, and an app will be used continually without end. This note supports the core insight, challenges several of its wordings, and offers a clearer argument. Information theory treats a message as novel insofar as it reduces uncertainty *given what the receiver already knows* [1]. Living organisms maintain local physical order by exporting thermodynamic entropy [2]; that is not the same quantity as Shannon information, nor is it a license to say people "reduce entropy" in an unrestricted physical sense. Predictive-processing accounts propose that brains minimize surprise relative to a model of a livable world, yet they also explore, because a dark, unchanging room would itself be surprising for creatures like us [3–5]. Novelty can enhance encoding into memory through midbrain dopamine and hippocampal circuits [6–8]. Rumination research, however, describes not an empty mind but a rigid, abstract, self-focused recycling of the same material, which persists partly because it interferes with concrete action [9–13]. Unbounded, unresolved novelty can create a second loop: an information-gap that never closes, refreshed on a variable schedule, so the app is used without end [14–18]. The refined claim is therefore double. Genuine, model-updating novelty is needed to interrupt a closed thought-loop. Hook-novelty that never resolves is itself a loop. Project Ariel should grow stories from the student's own canvas seed, refuse mere repetition, nudge from abstract rumination toward concrete rest and action, and offer an explicit "enough" ending.

In plain words

A mind that only hears itself can get stuck. A mind that is endlessly teased with newness can get stuck in a different way. What helps is a new connection that actually changes the pattern—then an honest place to stop.

Dedication
All credit and glory to God. I owe my knowledge to my Lord Jesus Christ.

01 · THE POSTULATE

What was claimed, and what we will test

The postulate, in Espinosa's words:

> *update the research for the need of novelty as entropy is novelty consumed to loose entropy for humans can reduce entropy, I postulate the release of novelty has to gain novelty for the brain if no new connections of entropic thoughts would just ruminate with no new connections the rummination might not stop and the app will be used continually without end*

Read generously, two complementary meanings are present. First, *consuming* novelty reduces uncertainty: the brain "spends" unpredicted information and becomes more ordered locally. That spending must be replenished, or thought has only itself to work with. Second, when an app *releases* novelty—when it tells a story, offers a prompt, or opens a canvas—that release must actually produce new connections for the student. If it does not, the same thoughts recycle, rumination may not stop, and the student keeps returning to the app in search of a resolution that never arrives.

A careful restatement, before any debate:

> If a person's current thought-pattern receives no model-updating input and forms no new connections among its parts, it can recycle as rumination and may not stop on its own. An app that fails to supply genuine novelty, or that supplies only unresolved surface novelty, can be used continually without end.

The rest of this note asks four questions. Which parts of that restatement are already supported? Where do the words "entropy" and "novelty" slide between different meanings? Does novelty alone stop rumination? And can the pursuit of novelty become the very compulsive use the postulate fears?

> **Novel contribution.** A bounded synthesis: distinguish thermodynamic entropy, information, and neural repertoire; hold together surprise-minimization and novelty-seeking; separate rumination-as-rigidity from rumination-as-emptiness; and derive four design rules for Ariel. This is a new argument within the project lineage, not a claim of priority over the cited sciences, and not a proof that the design works.

The verbatim wording is retained as provenance. Later sections refine it; they do not replace the author's voice with a quieter one.

02 · THREE DIFFERENT "ENTROPIES"

The most important correction

The postulate treats entropy, novelty, and the brain's ability to "reduce entropy" as nearly the same idea. Research supports a family resemblance, not an identity. Three quantities need to be kept apart.

Shannon information

Claude Shannon defined the information in a message as the reduction of uncertainty about which signal would arrive, given a known set of possibilities [1]. In that vocabulary, a stimulus is novel to a particular person insofar as it was not already predicted by that person's current model. The same sentence can be old news to one student and a genuine surprise to another. Consuming novelty, in this sense, means updating a model: after the update, uncertainty about *that* item has fallen. Shannon's entropy is dimensionless information, not heat, not mood, and not a moral quality.

Thermodynamic entropy

Erwin Schrödinger argued that living organisms remain ordered by continually drawing "negative entropy" from their surroundings—more plainly, by taking in free energy and exporting the disorder they produce [2]. A person can locally maintain a highly ordered body and a highly ordered thought, while the larger physical system, including the waste heat of metabolism, still obeys the second law. Humans do not repeal thermodynamic entropy. They rearrange it. This is the honest reading of "humans can reduce entropy": locally, and at a cost paid elsewhere.

Neural repertoire

Robin Carhart-Harris and colleagues use entropy for the breadth of a brain's transient connectivity motifs: how many different patterns it visits, and how hard those patterns are to predict [13]. On that account, ordinary waking thought is relatively constrained, and some difficult states—depression among them—are described as too ordered, too inflexible, rather than too chaotic. That is already a warning against saying rumination is simply "high entropy thought." If anything, the entropic-brain sketch places unyielding, stereotyped cognition toward the low-entropy, subcritical pole [13]. The postulate's fear that thoughts will "just ruminate with no new connections" is closer to this picture of rigidity than to a picture of formless chaos.

These three uses share a metaphor of order and uncertainty. They are not interchangeable measurements. Mixing them produces sentences that sound deeper than they are. The argument below therefore uses "novelty" for unpredicted, model-updating information; "uncertainty" for Shannon surprise relative to a person's current expectations; and "rigidity" for a thought-pattern that recycles without forming new connections. Physical entropy is mentioned only to keep the thermodynamic claim in its proper place.

Word	What it measures	What it does not measure
Shannon entropy / information	Uncertainty about a signal, given a model [1]	Heat, mood, or moral worth
Thermodynamic entropy	Physical disorder / unavailable energy; life exports it locally [2]	Whether a thought is wise or stuck
Neural repertoire	How many transient brain patterns are visited [13]	Whether those patterns are kind, true, or useful
Psychological novelty	That this person did not already predict this	That the stimulus is objectively new in the universe

Landauer's principle and related physics explore a deep link between information and thermodynamics. That link does not make a new story in an app equivalent to a change in bodily heat, and this paper does not rely on it.

03 · A BRAIN THAT HATES SURPRISE STILL SEEKS NOVELTY

The dark-room problem, and why it matters

Karl Friston's free-energy principle proposes that biological agents persist by minimizing surprise—or a tractable bound on surprise—relative to an internal model of the world [3]. Taken naively, that sounds like an argument for sitting in a dark room forever: the least surprising sensory stream is the one that never changes. Friston, Thornton, and Clark call this the dark-room problem, and they resolve it by noting that surprise is always surprise *given a model* [4]. Creatures like us have been shaped to expect a rich, viable world. For such a creature, a dark unchanging room is itself highly surprising. Average surprise is H(s | m), a function of sensations and the agent, not the raw variability of the sensory stream [4].

A related line of work asks how an agent that minimizes expected free energy can still explore. Schwartenbeck, FitzGerald, Dolan, and Friston argue that novelty and surprise-minimization are compatible when exploration is a way of improving the model and thereby reducing future uncertainty [5]. The brain does not seek novelty as an end in itself; it seeks the information that will make the world more predictable later.

That computational story has a dopaminergic counterpart. Kakade and Dayan proposed that dopamine cells multiplex ordinary reward-prediction errors with novelty-related "bonuses," including exploration bonuses, that encourage an animal to visit new states [6]. Bunzeck and Düzel found that the human substantia nigra / ventral tegmental area responded to absolute stimulus novelty rather than to other kinds of salience, and that a novel context enhanced learning even for familiar items presented alongside it [7]. Lisman and Grace described a hippocampal–VTA loop in which hippocampal novelty signals help control what is admitted into long-term memory [8]. In plain words: novelty is not merely entertaining. Under the right conditions it is the door through which a new connection is allowed to form.

This is the strongest support for the postulate's first half. If no unpredicted information arrives, the model has nothing new to encode. Hippocampal and midbrain novelty responses stay quiet. Thought is left to recombine what it already has. The "need of novelty" is real in that specific sense: learning and flexible thought depend on model-updating input, not on stimulation as such.

Two limits follow immediately. First, the dark-room resolution also implies its inverse, which we might call a bright-room problem: an environment of endless, unstructured surprise is not what a free-energy agent wants either. Optimal exploration is bounded [5,15]. Second, novelty that does not enter memory—novelty that is noticed and discarded—does not create the "new connections" the postulate requires. The hippocampal–VTA loop is about encoding, not about keeping the eyes busy [8].

The free-energy principle is a unifying proposal, not a settled law, and this paper does not treat Ariel as an implementation of it. The design claim uses only the narrower, better-supported point: people both reduce uncertainty and explore in order to reduce later uncertainty.

04 · RUMINATION IS RIGIDITY, NOT SILENCE

Where the postulate is right, and where the mechanism changes

Susan Nolen-Hoeksema's response-styles theory proposed that people who respond to low mood by repeatedly focusing on their symptoms, and on the causes and consequences of those symptoms, remain depressed longer than people who turn to instrumental activity [9]. A later review with Wisco and Lyubomirsky kept the core claim and sharpened it:

rumination is perseverative, self-focused, and difficult to disengage from; it impairs problem-solving and interferes with the very actions that might change the situation [10].

Edward Watkins showed that repetitive thought is not always harmful. What matters includes the valence of the content, the context, and—especially—the level of construal [11]. Abstract repetitive thought ("Why am I like this? What does this say about me?") tends to be unconstructive. Concrete repetitive thought ("What exactly happened? What is one next step?") can support planning and recovery [11]. This is a direct challenge to any design that answers rumination with more open-ended self-inquiry of the same abstract kind.

At the level of large-scale brain networks, Hamilton, Farmer, Fogelman, and Gotlib reviewed evidence linking depressive rumination to altered coupling between the default-mode network, which supports self-referential thought, and affectively laden medial prefrontal regions [12]. Carhart-Harris and colleagues, citing related findings, describe depressed cognition as inflexible and overly fixed, a state in which one introspective mode comes to dominate [13]. The problem is not that the mind has gone quiet. It is that a particular, self-focused pattern has become too easy to re-enter and too hard to leave.

So the postulate is right that rumination can continue "without new connections" and "might not stop." It is less right if "no new connections" is heard as an empty or high-entropy mind. Rumination is often densely connected—but connected in a loop, at an abstract level, about the self. The missing ingredient is not activity. It is a new kind of connection: one that reaches outside the loop, preferably into something concrete and present.

That is why behavioral activation matters for this debate. In a randomized trial, Dimidjian and colleagues found that helping people re-engage with concrete, rewarding activity was comparable to antidepressant medication among more severely depressed adults, and compared favorably with cognitive therapy in that group [17]. The intervention does not work by flooding the person with novel stimuli. It works by restoring contact with the world through specific acts. Combined with Watkins's abstract/concrete distinction [11] and Nolen-Hoeksema's claim that rumination interferes with instrumental behavior [9,10], the implication is clear: novelty without a path into action is not a sufficient answer to rumination.

Boredom research adds a further caution. Eastwood, Frischen, Fenske, and Smilek define boredom as an aversive state in which a person wants to be engaged but cannot successfully direct attention to internal or external information [16]. A bored or ruminating student may reach for an app not because it offers a new connection, but because it promises relief from an unengaged mind. If the app then supplies only more unstructured inwardness, or only flickering stimulation, both the boredom and the rumination can persist.

> **Refined mechanism.** Rumination persists when thought stays abstract, self-focused, and cut off from new encoding and from concrete action. The needed "new connections" are not more thoughts of the same kind. They are connections that update the model or return the person to a particular, present act.

None of these sources validate an app-based treatment. Behavioral activation in [17] was delivered as a clinical intervention, not as a story generator. Ariel may borrow the principle of concreteness; it must not claim the trial's results.

05 · WHEN NOVELTY BECOMES THE LOOP

The second failure mode

The postulate warns that if novelty fails, "the app will be used continually without end." That prediction can be true for two opposite reasons, and the design must answer both.

George Loewenstein interpreted curiosity as a cognitively induced deprivation: a person notices a gap in what they know and is motivated to close it [14]. Daniel Berlyne had already argued that organisms seek an intermediate level of novelty and complexity—enough to arouse, not so much as to overwhelm [15]. Together these accounts explain why a well-judged story can feel like a drink of water, and why an endless sequence of half-answered hooks can feel like thirst that is never quenched. An information-gap that is repeatedly opened and never closed will keep calling the student back [14].

Product research has documented how that psychology is used. Montag, Lachmann, Herrlich, and Zweig review addictive features of social-media platforms and freemium games against psychological and economic theories, including intermittent or variable rewards, social-comparison pressures, and designs that remove a natural stopping point [18]. Variable reward is powerful precisely because the next item might be the satisfying one. The student is not nourished by novelty. The student is kept adjacent to it.

This is the bright-room problem in commercial form. Dopamine novelty bonuses [6,7], which evolved to support exploration and encoding [8], can be captured by a feed that is always slightly unpredicted and never complete. The app is then used "continually without end," not because it failed to offer stimulation, but because it offered the wrong kind: surface novelty that does not update the model, close the gap, or end.

Berlyne's optimal-arousal idea also warns against the opposite error of flooding the student with maximal randomness [15]. Pure randomness is novel in a Shannon sense and meaningless in a human one. A random canvas mark is a seed, not a story. Meaning requires a gentle architecture—character, a small longing, a return—that can turn unpredicted marks into a connection the student can remember. That architecture is already a form of entropy suppression in Carhart-Harris's vocabulary: enough constraint for a coherent experience, enough openness for surprise [13].

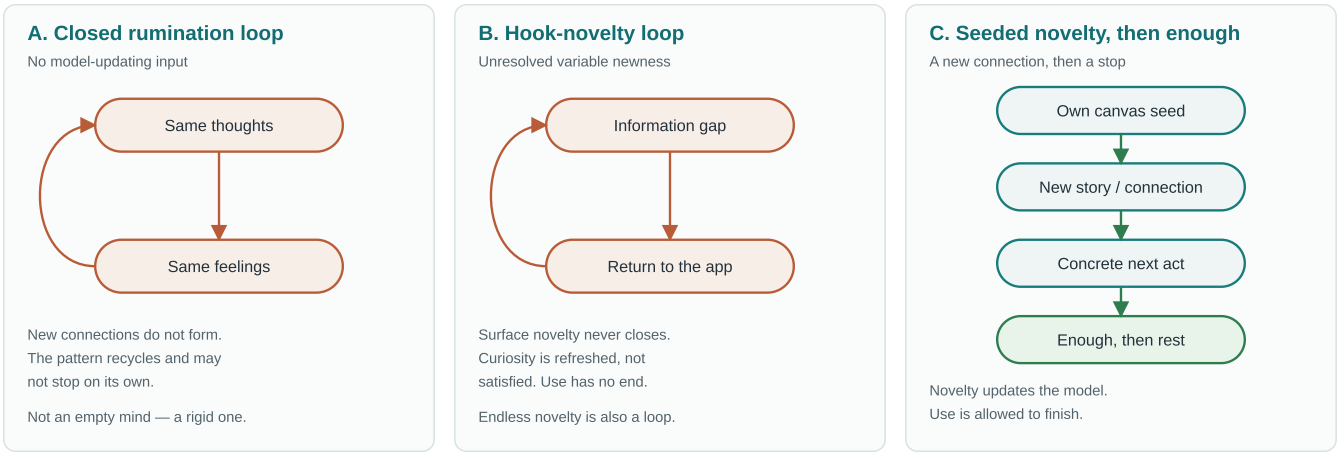


Figure 1. Two ways an app can fail, and one proposed path. Panel A is the postulate's first fear: thought recycles without new connections. Panel B is the second fear, refined: unresolved novelty becomes its own compulsion. Panel C is the design response: novelty grown from the student's own seed, turned toward a concrete act, and allowed to finish.

Figure 1 is a conceptual diagram, not a measured pathway and not a clinical flowchart.

06 · THE REFINED ARGUMENT

Premises, conclusion, and predictions that can fail

The debate can now be gathered into a single argument. Each premise is labeled by the kind of support it has.

1. **Information (well established as a definition).** A stimulus is novel to a person insofar as it reduces uncertainty in that person's current model [1]. Consuming novelty means updating the model, not merely encountering new pixels.
2. **Local order (well established in outline).** Organisms maintain local physical order by exporting thermodynamic entropy [2]. This supports a careful reading of "humans can reduce entropy," and forbids an unrestricted one.
3. **Exploration in the service of prediction (influential theory, partial evidence).** Brains minimize surprise relative to a model of a livable world and seek novelty when doing so improves that model [3–5]. Midbrain and hippocampal systems can treat novelty as a signal that something is worth encoding [6–8].
4. **Rumination as rigidity (well supported as a clinical-psychological construct).** Unconstructive rumination is repetitive, abstract, and self-focused. It persists partly because it blocks instrumental action, and it has been associated with relatively constrained, self-referential network dynamics [9–13]. The missing "new connections" are new encodings and concrete acts, not more of the same inward talk.
5. **Two loops, not one (the postulate, refined).** Absence of model-updating input favors recycling of existing thought (Figure 1A). Presence of unresolved, variable, hook-style novelty favors compulsive return without resolution (Figure 1B) [14–16,18]. Either failure can make an app used continually without end.
6. **Action and an ending (well supported outside this app; untested inside it).** Shifts from abstract to concrete processing, and re-engagement with specific activity, have better credentials as rumination-interrupters than unbounded new stimuli [11,17]. An explicit stopping point is the opposite of the no-stop designs reviewed by Montag and colleagues [18].

Conclusion. Project Ariel should supply genuine, model-updating novelty so that thought is not left to ruminate on itself; derive that novelty, where possible, from the student's own act, so it is personally unpredicted yet meaningful; refuse to serve mere repetition back as newness; interrupt abstract looping with a restful, concrete turn; and include an "enough" state so that use is allowed to end.

Predictions that can fail

These are design hypotheses, not results. A null or reversed finding should be reported rather than explained away after the fact.

1. Holding time on task constant, a story grown from the student's own canvas seed should reduce subsequent self-reported abstract rumination more than a repeated generic prompt. If the two conditions do not differ, personal seeding is not doing the work claimed here.
2. A novelty check that maps the same seed to the same story, and declines near-duplicate stories, should reduce immediate re-opening of the app compared with a condition that varies similar content on each visit. If variable near-duplicates produce less re-opening, the hook-loop concern is weakened.
3. An explicit "enough" ending plus a concrete rest nudge should shorten sessions and reduce next-day compulsive return compared with an endless-feed control, without increasing rumination. If the ending increases distress or rumination, the stop is premature or poorly worded.
4. Raising novelty while keeping the content abstract and self-focused ("here are more questions about why you are like this") should *not* reliably reduce rumination. If it does, then novelty-as-such is doing more work than this argument allows, and premise 4 needs revision.
5. After a seeded story, one concrete next-act prompt should outperform an extra novel story on later rumination. This is the behavioral-activation and concreteness prediction [11,17] transferred into Ariel's setting. If the extra story wins, the argument has overstated the need for an exit into action.

A modest observational check—not a treatment trial—could compare these conditions in consenting adults using local, skippable, nonclinical ratings of abstract versus concrete thought, session length, and voluntary return. Ethics review, an accessible distress protocol, and a clear statement that the app is not care would be required before any such study. No claim is made that these predictions have already been tested.

Sample size, exact rating items, and analysis plans would need to be fixed in advance of seeing results. This paper does not offer them.

07 · WHAT THIS MEANS FOR ARIEL

Four design rules, and a theological caution

The immediate purpose of this note is not a new equation. It is a set of constraints on the Create tab and on the wider suite, so that novelty serves discovery rather than proving, and so that use has an end.

1. Novelty from the student's own canvas seed

The student's marks are a source of unpredicted structure. A seed derived from those marks can grow a story nobody has heard before, including the student, without scraping the world for optimized bait. The same drawing should always tell the same story: determinism is what makes the novelty honest. If the marks did not change, nothing new needed to be invented. This is closer to uncovering something already latent in the student's act than to manufacturing surprise. It also respects Shannon's point that novelty is relative to a model [1] and Lisman and Grace's point that novelty matters when it can be encoded [8].

2. A novelty check against repeats

Ariel should notice when it is about to serve a near-duplicate of a recent story, prompt, or lesson and should decline to dress that repeat up as discovery. The accomplishment record already exists so the student can remember; it should not become a slot machine of slightly reworded praise. A repeat offered as a remembered stone is different from a repeat offered as news.

3. An anti-rumination rest nudge

If the student's language stays abstract, self-focused, and circular—proving, explaining the self, asking "why am I like this?"—Ariel should not answer with more of the same. The kinder move is the one Watkins and the behavioral-activation literature would recognize: come back to something particular and present [11,17]. A rest card already used in the suite ("This sounds like it might be about proving. Want to come back to wonder?") sits in this family. The nudge is an invitation, never a verdict and never a diagnosis.

4. An "enough" state, so use can end

The suite's enough-state language—"You've learned what you needed here. Nothing more to prove. And there is always more to enjoy."—is not a flourish. On this argument it is a safety feature. It is the opposite of a feed with no stopping point [18]. Grace both starts and ends the larger loop the project has described elsewhere. An app that cannot say "this is enough" will, exactly as the postulate warns, be used continually without end.

Failure mode	What it feels like	Ariel's answer
Closed rumination (Fig. 1A)	The same thoughts, no new connection	A seeded story; then a concrete, present act
Hook novelty (Fig. 1B)	Almost satisfied, never finished	Deterministic seeds; no near-duplicate "newness"; an enough state
Randomness without meaning	New, but empty	A gentle story shape around the student's marks
Abstract self-inquiry	More explaining, more proving	A rest nudge toward wonder and the particular

A theological note, held lightly

Scripture already knows both sides of this argument. Ecclesiastes warns that "the eye is not satisfied with seeing, nor the ear filled with hearing," and that there is "no new thing under the sun" (Ecclesiastes 1:8–10, KJV). That is the hook-loop, spoken from a long time ago: consumption of novelty does not satisfy. Lamentations answers with a different kind of newness: "It is of the Lord's mercies that we are not consumed, because his compassions fail not. They are new every morning: great is thy faithfulness" (Lamentations 3:22–23, KJV). Isaiah hears, "Behold, I will do a new thing" (Isaiah 43:19, KJV), and the one seated on the throne says, "Behold, I make all things new" (Revelation 21:5, KJV).

The distinction is motive and source. Babel's fault, as this project has said elsewhere, was "let us make us a name." Endless consumed novelty is a small Babel: newness pursued so the self can keep going. Morning mercies are received, not mined. They do not require the student to prove anything, and they do not require the app to stay open. In that light, the canvas seed is less like a slot machine and more like a sunflower: the particular mark is the student's, but the architecture that can turn a seed into something living was already there. This is faith language, not an experimentally tested mechanism. Participants and readers remain free to take or leave it.

> **Safety.** This note is not crisis care, and Project Ariel is not a treatment for rumination, depression, or compulsive use. In the U.S., call or text **988** for crisis support. For immediate danger or a medical emergency, call **911**. Do not delay professional care to complete an app exercise or to finish a story.

08 · LIMITS, REFERENCES, AND PROVENANCE

What this paper does not prove

No human study was conducted. No brain was imaged, no rumination scale was administered, and no A/B test of Ariel was run. The citations establish that the named papers and books exist and that their accessible abstracts, records, or full texts support the limited background claims attached to them. They do not validate Ariel's interface, the canvas-seed method, the enough state, or any coefficient from earlier MindFrame work.

Several further limits should be stated plainly. The free-energy principle and the entropic-brain hypothesis are active, contested research programs, not textbooks facts [3,13]. Default-mode findings in depression are heterogeneous; Hamilton and colleagues themselves treat the mapping as a model under construction [12]. Behavioral activation's clinical results [17] do not transfer automatically to a brief digital story. Self-reported rumination is not the same as a neural entropy measure, and neither is a moral evaluation of a person's thought. Individual differences in novelty-

seeking, trauma, attention, culture, and faith will shape how any of these ideas feel in practice. More novelty is not always kinder: for some people, unexpected content can startle or overwhelm, which is one reason Berlyne placed an optimum rather than a maximum on arousal [15].

The three strongest challenges to the original wording are therefore retained, not hidden:

- **Entropy is not one thing.** Thermodynamic entropy, Shannon information, and neural repertoire share a family resemblance and must not be treated as a single dial the app can turn.
- **Novelty alone does not stop rumination.** Concrete processing and contact with particular action have better evidence than stimulation as such.
- **Endless novelty can become its own compulsion.** An app used without end may be failing by teasing newness, not only by withholding it.

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Citation scope: bibliographic records and accessible abstracts or full texts were checked on 10 November 2026. These sources establish the existence and general orientation of the cited work. None validates this paper's design rules, predictions, or any Ariel interface. Scriptural quotations are from the King James Version and are used as theological conversation, not as empirical evidence.

Project provenance and reproducibility

Espinosa, R. (2026). MindFrame and Project Ariel, evolving personal conceptual projects. Context supplied by the creator. Related notes in this series include *After the Line*, *The Void-State Framework*, *The Gated Line*, and the Project Ariel Students papers. Repository identified by the creator: github.com/ryryespi/mindframe. Public research index: ryryespi.github.io/mindframe/research/.

Grok Bot, an AI assistant, helped restate the postulate, search and verify citation records, draft the argument, draw Figure 1, and render this document. No human research has been performed. Files: paper.html, loops.svg, and the accompanying PDF.

Conclusion: the need of novelty is real, but novelty must gain a new connection and then know how to stop. The next honest step is to build those four rules into Create—and, later, to let the predictions fail if they are going to fail—not to claim that rumination has been solved.